

Rajiv Khanna

Curriculum Vitae

September 2026

Address: Department of Computer Science
Purdue University,
West Lafayette, USA.
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Research Interests

Machine Learning Theory: Provable guarantees for optimization and generalization.

Applications: Demystifying Large Scale Neural Networks through Interpretability.

Education

University of California, Berkeley

Postdoctoral Researcher Advisor: Prof. Michael Mahoney, Jan 2019 - May 2021

University of Texas at Austin, Austin TX

PhD in Electrical and Computer Engineering (Advisor: Prof. Joydeep Ghosh), Aug 2018.

Indian Institute of Technology Bombay, Mumbai India

Masters in Computer Science and Engineering (Advisor: Prof. Soumen Chakrabarti), July 2008.

National Institute of Technology Jalandhar India

Bachelors in Computer Science and Engineering July 2006.

Appointments

Assistant Professor

Purdue University, Jan 2022 - Now

Visiting Researcher

Google, Aug 2021 - Dec 2021

Research Fellow

Simons Institute for the Theory of Computing, UC Berkeley, Aug 2018 - Dec 2018

Research Assistant

University of Texas at Austin, Aug 2012 - Aug 2018

Teaching Assistant

University of Texas at Austin, Aug 2013 - Aug 2014

Research Engineer

Yahoo! Labs, Aug 2008 - Jul 2012

Honors

NSF CAREER AWARD 2026

AnalytiXin Fellowship 2023

Best paper award, Neurips 2020.

Research Fellowship, Simons Institute for the Theory of Computing, UC Berkeley, 2018.

Most outstanding student of batch 2008, IIT Bombay 2008

Research Impact

Number of citations (from Google Scholar) > 2900.

H-index (from Google Scholar): 20

i10-index (from Google Scholar): 25

Technical reports/working papers

1. Behdin, K., Q. Song, A. Gupta, S. Keerthi, A. Acharya, B. Ocejjo, G. Dexter, R. Khanna, D. Durfee, and R. Mazumder (2023). msam: Micro-batch-averaged sharpness-aware minimization. *arXiv preprint arXiv:2302.09693*.
2. Dexter, G., R. Khanna, J. Raheel, and P. Drineas (2023). Feature space sketching for logistic regression. *arXiv preprint arXiv:2303.14284*.
3. Keerthi, S. S., T. Schnabel, and R. Khanna (2015). Towards a better understanding of predict and count models. *arXiv preprint arXiv:1511.02024*.
4. Park, Y., R. Khanna, J. Ghosh, and D. Mihalko (2015). DPM: A State Space Model for Large-Scale Direct Marketing. *arXiv preprint arXiv:1507.01135*.

Peer-reviewed workshops

1. Kim, Y. I., P. Agrawal, J. O. Royset, and R. Khanna (2023). On Memorization and Privacy risks of Sharpness Aware Minimization. *ICML 2023 (Workshop on Data-centric Machine Learning Research)*.
2. Guha, S., R. Khanna, and S. Koyejo (2021). Distribution Preserving Bayesian Coresets using Set Constraints. In: *NeurIPS Workshop Distribution shifts: connecting methods and applications (DistShift)*.

Journals

1. Elenberg, E. R., R. Khanna, A. G. Dimakis, and S. Negahban (2018). Restricted strong convexity implies weak submodularity. *The Annals of Statistics* **46**(6B), 3539–3568.

Conferences

1. Khanna, R. (2026). Spectral Valleys and Sharp Failures in Greedy Determinant Maximization. *COLT 2026*.
2. Kim, Y. I., P. Agrawal, A. Agiollo, J. O. Royset, and R. Khanna (2026). On the Privacy Risks of Sharpness Aware Minimization. *ICLR 2026*.
3. Mahmood, S. H. A., Y. Gao, Y. Souiri, S. Som, M. Yin, R. Khanna, and X. Song (2026). Consistent Diffusion Language Models. *ICML 2026*.
4. Mahmood, S. H. A., H. Y. Tian, X. Duan, C.-J. Ho, R. Khanna, and M. Yin (2026). From Fallback to Frontline: When Can LLMs be Superior Annotators of Human Perspectives? *ACL (Findings) 2026*.
5. Mahmood, S. H. A., M. Yin, and R. Khanna (2026). Align When They Want, Complement When They Need! Human-centered Ensembles for Adaptive Human-AI Collaboration. *(AAAI 2026)*.
6. Tang, H. and R. Khanna (2026). Sharpness-Aware Machine Unlearning. *ICLR 2026*.
7. Chang, W.-k. and R. Khanna (2025a). A Unified Stability Analysis of SAM vs SGD: Role of Data Coherence and Emergence of Simplicity Bias. *Neurips 2025*.
8. Chang, W.-k. and R. Khanna (2025b). Stable Coresets via Posterior Sampling: Aligning Induced and Full Loss Landscapes. *Neurips 2025*.
9. He, K., P. Drineas, and R. Khanna (2025). Structure-Aware Spectral Sparsification via Uniform Edge Sampling. *Neurips*.
10. Mahmood, S. H. A., M. Yin, and R. Khanna (2025a). On the Support Vector Effect in DNNs: Rethinking Data Selection and Attribution. In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, pp.1020–1031.
11. Agiollo, A., Y. I. Kim, and R. Khanna (2024). Approximating Memorization Using Loss Surface Geometry for Dataset Pruning and Summarization. In: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp.17–28.
12. Dexter, G., P. Drineas, and R. Khanna (2024). The Space Complexity of Approximating Logistic Loss. In: *Advances in Neural Information Processing Systems*.
13. Dexter, G., B. Ocejjo, S. Keerthi, A. Gupta, A. Acharya, and R. Khanna (2024). A precise characterization of sgd stability using loss surface geometry. *International Conference on Learning Representations (ICLR)*.
14. Quinzan, F., R. Khanna, M. Hershcovitch, S. Cohen, D. G. Waddington, T. Friedrich, and M. W. Mahoney (2023). Fast feature selection with fairness constraints. *AISTATS*.
15. Sachs, S., T. van Erven, L. Hodgkinson, R. Khanna, and U. Şimşekli (2023). Generalization Guarantees via Algorithm-dependent Rademacher Complexity. In: *The Thirty Sixth Annual Conference on Learning Theory*. PMLR, pp.4863–4880.

16. Hodgkinson, L., U. Simsekli, R. Khanna, and M. Mahoney (2022). Generalization bounds using lower tail exponents in stochastic optimizers. In: *International Conference on Machine Learning*. PMLR, pp.8774–8795.
17. Gupta, V., A. Ghosh, M. Dereziński, R. Khanna, K. Ramchandran, and M. W. Mahoney (2021). LocalNewton: Reducing communication rounds for distributed learning. In: *Uncertainty in artificial intelligence*. PMLR, pp.632–642.
18. Utrera, F., E. Kravitz, N. B. Erichson, R. Khanna, and M. W. Mahoney (2021). Adversarially-trained deep nets transfer better: Illustration on image classification. *International Conference on Learning Representations*.
19. Zhang, J., R. Khanna, A. Kyrillidis, and S. Koyejo (2021). Bayesian coresets: Revisiting the nonconvex optimization perspective. In: *International Conference on Artificial Intelligence and Statistics*. PMLR, pp.2782–2790.
20. Dereziński, M., R. Khanna, and M. W. Mahoney (2020). Improved guarantees and a multiple-descent curve for column subset selection and the nystrom method. *Advances in Neural Information Processing Systems* **33**, 4953–4964.
21. Yang, Y., R. Khanna, Y. Yu, A. Gholami, K. Keutzer, J. E. Gonzalez, K. Ramchandran, and M. W. Mahoney (2020). Boundary thickness and robustness in learning models. *Advances in Neural Information Processing Systems* **33**, 6223–6234.
22. Khanna, R., B. Kim, J. Ghosh, and S. Koyejo (2019). Interpreting black box predictions using fisher kernels. In: *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, pp.3382–3390.
23. Zhang, J. Y., R. Khanna, A. Kyrillidis, and O. O. Koyejo (2019). Learning sparse distributions using iterative hard thresholding. *Advances in Neural Information Processing Systems* **32**.
24. Joshi, S., R. Khanna, and J. Ghosh (2018). Co-regularized Monotone Retargeting for Semi-supervised LeTOR. In: *Proceedings of the 2018 SIAM International Conference on Data Mining*. Society for Industrial and Applied Mathematics, pp.432–440.
25. Khanna, R. and A. Kyrillidis (2018). IHT dies hard: Provable accelerated iterative hard thresholding. In: *International Conference on Artificial Intelligence and Statistics*. PMLR, pp.188–198.
26. Locatello, F., G. Dresdner, R. Khanna, I. Valera, and G. Rätsch (2018). Boosting black box variational inference. *Advances in Neural Information Processing Systems*.
27. Locatello, F., R. Khanna, J. Ghosh, and G. Ratsch (2018). Boosting variational inference: an optimization perspective. In: *International Conference on Artificial Intelligence and Statistics*. PMLR, pp.464–472.
28. Khanna, R., E. Elenberg, A. Dimakis, S. Negahban, and J. Ghosh (2017). Scalable greedy feature selection via weak submodularity. In: *Artificial Intelligence and Statistics*. PMLR, pp.1560–1568.
29. Khanna, R., E. R. Elenberg, A. G. Dimakis, J. Ghosh, and S. Negahban (2017). On approximation guarantees for greedy low rank optimization. In: *International Conference on Machine Learning*. PMLR, pp.1837–1846.
30. Khanna, R., J. Ghosh, R. Poldrack, and O. Koyejo (2017a). Information projection and approximate inference for structured sparse variables. In: *Artificial Intelligence and Statistics*. PMLR, pp.1358–1366.
31. Khanna, R., J. Ghosh, R. Poldrack, and O. Koyejo (2017b). A deflation method for structured probabilistic PCA. In: *Proceedings of the 2017 SIAM International Conference on Data Mining*. Society for Industrial and Applied Mathematics, pp.534–542.
32. Locatello, F., R. Khanna, M. Tschannen, and M. Jaggi (2017). A unified optimization view on generalized matching pursuit and frank-wolfe. In: *Artificial Intelligence and Statistics*. PMLR, pp.860–868.
33. Khanna, R., M. Tschannen, and M. Jaggi (2016). Pursuits in Structured Non-Convex Matrix Factorizations. *arXiv preprint arXiv:1602.04208*.
34. Kim, B., R. Khanna, and O. O. Koyejo (2016). Examples are not enough, learn to criticize! criticism for interpretability. In: *Advances in Neural Information Processing Systems*, pp.2280–2288.
35. Khanna, R., J. Ghosh, R. Poldrack, and O. Koyejo (2015). Sparse submodular probabilistic PCA. In: *Artificial Intelligence and Statistics*. PMLR, pp.453–461.
36. Koyejo, O. O., R. Khanna, J. Ghosh, and R. Poldrack (2014). On prior distributions and approximate inference for structured variables. *Advances in Neural Information Processing Systems* **27**.
37. Khanna, R., L. Zhang, D. Agarwal, and B.-C. Chen (2013). Parallel matrix factorization for binary response. In: *2013 IEEE International Conference on Big Data*. IEEE, pp.430–438.

38. Agarwal, D., R. Agrawal, R. Khanna, and N. Kota (2010). Estimating rates of rare events with multiple hierarchies through scalable log-linear models. In: *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*, pp.213–222.
39. Agarwal, D., E. Gabrilovich, R. Hall, V. Josifovski, and R. Khanna (2009). Translating relevance scores to probabilities for contextual advertising. In: *Proceedings of the 18th ACM conference on Information and knowledge management*, pp.1899–1902.
40. Chakrabarti, S., R. Khanna, U. Sawant, and C. Bhattacharyya (2008). Structured learning for non-smooth ranking losses. In: *Proceedings of the 14th ACM SIGKDD international conference on knowledge discovery and data mining*, pp.88–96.

Funding

AnalytIXIN Fellowship (Unrestricted gift Jan 2023-Dec 2025) from Central Indiana Corporate Partnership - 200K.

NSF CAREER AWARD 2026. \$600k

Teaching

Spring 2022 Interpretability in Machine Learning (Seminar)

Fall 2022 CS573 Data Mining and Machine Learning (Graduate)

Spring 2023 CS573 Data Mining and Machine Learning (Graduate)

Fall 2023 CS373 Data Mining (Undergraduate)

Spring 2024 CS573 Data Mining and Machine Learning (Graduate)

Spring 2025 CS573 Data Mining and Machine Learning (Graduate)

Fall 2025 CS573 Data Mining and Machine Learning (Graduate)

PhD Students

Haoran Tang (PhD 2st year) [TK26]

Kaiwen He (PhD 3rd year) (with Petros Drineas) [HDK25]

Wei-kai Chang (PhD 3rd year) [CK25a; CK25b]

Young In Kim (PhD 4th year) [AKK24; Kim+23]

Syed Hasan Mahmood (PhD 4th year) (with Ming Yin) [MYK25a; MYK26; MYK25b]

Graduated

Gregory Dexter PhD May 2024 (joint with Petros Drineas), LinkedIn [Beh+23; Dex+24; Dex+23; DDK24]

Invited Talks

A Unified Stability Analysis of SAM vs SGD: Role of Data Coherence and Emergence of Simplicity Bias. *Symposium on Understanding Double Descent at SIAM Mathematics of Data Science*, Jul 2025

Shape of Learning: Building Structure-Aware Systems That Understand Their Data. *Amazon Headquarters, Seattle*, Jul 2025

Characterizing SGD Stability Using Loss Surface Geometry. *Symposium on Understanding Double Descent at SIAM Mathematics of Data Science*, Oct 2024

Data Pruning using Loss Surface Geometry. *International Conference on AI and ML*, June 2024

Interpretability: Theory and Applications. *Fourth Indian Symposium on Machine Learning (IN-DOML)*, Indian Institute of Technology Bombay, Dec 2023

Theory of Feature Selection. *Robustness and Resilience in Stochastic Optimization and Statistical Learning: Mathematical Foundations* Ettore Majorana Foundation and Center for Scientific Culture, Erice, Italy May 2022

Academic Service

Purdue CS Purdue PhD Admission Committee for admissions in the Fall of 2023,2024,2025.

Area Chair/Senior PC: ICLR 2026, ICML2025, ICLR 2025, IJCAI 2025, IJCAI 2024, IJCAI 2023

Reviewer: Neurips (2016,2017,2018,2019,2020,2021,2022,2023,2024, 2025), ICML (2016,2017,2018,2019,2020,2021,2022,2023,2024), TMLR (2022,2023,2024), ALT 2024, AIS-TATS (2016,2017,2018,2019,2020,2021,2022,2023,2024), UAI (2020,2021,2022,2023,2024)